MIL-W-45562B
21 February 1984
SUPERSEDING
MIL-W-45562B
25 October 1973

# MILITARY SPECIFICATION

# WELDING AND SOLDERING EQUIPMENT, SUPPLIES, AND ACCESSORIES, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense

## 1 SCOPE

1.1 This specification covers the requirements for materials, methods of cleaning, drying, preservative applications, packing, and marking of welding and soldering equipment, supplies, and accessories for the various levels of protection for storage, shipping, and handling.

## 2 APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| O-S-801 | - Sulfuric Acid, Electrolyte; For Storage Batteries |
| QQ-S-781 | - Strapping, Steel and Seals |
| UU-S-48 | - Sacks, Shipping, Paper |
| UU-T-81 | - Tags, Shipping and Stock |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC PACK

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

PPP-B-15 - Bags and Envelopes, Cellophane, For Packaging
PPP-B-35 - Bags, Textile-Shipping, Burlap, Cotton and Waterproof Lamina'
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-576 - Boxes, Wood-Cleated, Veneer, Paper Overlaid
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-676 - Boxes, Setup
PPP-B-1364 - Box, Corrugated Fiberboard, High-Strength, Weather, Resistant, Double-Wall
PPP-C-96 - Cans, Metal, 28 Gage and Lighter
PPP-C-795 - Cushioning Material, Flexible, Cellular, Plastic Film For Packaging Applications
PPP-C-843 - Cushioning Material, Cellulosic
PPP-D-723 - Drums, Fiber
PPP-P-291 - Paperboard, Wrapping and Cushioning
PPP-T-42 - Tape, Pressure-Sensitive Adhesive; (General Packaging Application)
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 - Preservation-Packaging, Methods of
MIL-B-117 - Bags, Sleeves, and Tubing, Interior Packaging
MIL-B-121 - Barrier Material, Greaseproofed; Waterproofed, Flexible
MIL-B-208 - Battery, Storage, Lead Acid, Automotive and Navy Portable (Except Aircraft), Packaging and Packing of
MIL-P-775 - Hose, Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of
MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-pound Capacity
MIL-C-3955 - Cans, Fiber, Spirally Wound
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal
MIL-I-8574 - Inhibitors, Corrosion, Volatile, Utilization of
MIL-E-10062 - Engines: Preparation for Shipment and Storage of
MIL-W-10430 - Welding Rods And Electrodes; Preparation For Delivery Of
MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts; Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In
MIL-F-22019 - Film, Transparent, Flexible, Heat Sealable, Volatile Corrosion Inhibitor Treated


Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

MIL-B-22020 - Bags, Transparent, Flexible, Sealable, Volatile Corrosion Inhibitor Treated
MIL-B-26701 - Bottles, Screw Cap and Carboys Polyethylene Plastic
MIL-B-43014 - Boxes: Water Resistant Paperboard; Folding, Set-up, and Metal-Stayed
MIL-C-52950 - Crates, Wood, Open and Covered

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of
MIL-STD-794 - Parts and Equipment Procedures For Packaging and Packing of
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Industry association specifications and standards are generally available for reference from Libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one completed pack for first article inspection in accordance with 4.2.1. The first article may be either a preproduction pack or an initial production pack which conforms to all the requirements of this specification. In either case, the first article shall be identical with the production pack in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

3.2 General. Welding and soldering equipment, supplies, and accessories shall be preserved, packed, and marked in accordance with this specification or other specifications or standards as described herein. Specified requirements are assigned on the basis of the type of unit protection most commonly required for a specific item or equipment. Methods of preservation in general are based on susceptibility of the item to corrosion or other forms of deterioration, adaptability to the use of a contact preservative, weight, size, or need for mechanical or physical protection. Unless otherwise specified, when an item or equipment being procured in conformance with a commodity specification has packaging requirements which differ from this specification, the requirements of the commodity specification shall apply.

3.3 Materials. Materials used in preservation and packing shall be as specified herein and in applicable specifications or standards. Materials not specifically designated shall be of good commercial quality suitable for the purpose intended and subject to approval by the procuring activity. Materials shall be new and free from all defects and imperfections which may affect the serviceability of the completed package or packed product.

3.4 Disassembly and matchmarking. Disassembly shall be in accordance with MIL-STD-1186. Matchmarking shall be on tags conforming to UU-T-81, type A, class 2; size and marking of tags shall be in accordance with the requirements of MIL-STD-129.

3.5 Cleaning and drying. Unless otherwise specified (see 6.2), all surfaces of the item(s) shall be cleaned by one process or a combination of processes listed in MIL-P-116, and drying shall be accomplished through one or more drying procedures listed therein. Special cleaning and drying procedures may be employed upon approval by the procuring activity.

* 3.6 Preservation. Preservation shall be level A, B, or Industrial, as specified (see 6.2).

3.6.1 Level A.

3.6.1.1 Preservation (methods of). Preservative types and methods of preservation indicated by symbols herein shall conform to the requirements for the corresponding symbols in MIL-P-116. Unless otherwise specified (see 6.2), the preservative type and preservation method shall be as specified in Table I herein. Application of preservatives shall be in accordance with the applicability and guidelines of MIL-P-116. When a preservative is required, but not specified, the selection shall be at the option of the supplier.

3.6.1.1.1 Volatile corrosion inhibitors (VCI, P-18). When specified (see 6.2), VCI treated materials may be used as an alternate to the preservative indicated in column 3 of Table I. The utilization of volatile corrosion inhibitors shall be in accordance with MIL-I-8574. Unless otherwise specified (see 6.2), application of a contact preservative with VCI will not be required. Transparent, flexible, heat sealable VCI treated films and bags shall conform to MIL-F-22019 and MIL-B-22020 as applicable.

3.6.1.1.2 External metal surfaces. Critical, ferrous metal surfaces, other than stainless steel, shall be coated with type P-2 or P-11 preservative and over-wrapped with barrier material conforming to MIL-B-121, grade A. Barrier material shall be secured with tape conforming to PPP-T-60 (Type IV). Noncritical ferrous metal surfaces, not painted or plated, shall be coated with type P-1 or P-19 preservative and an overwrap is not required. Completely painted items shall be processed method III.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

3.6.1.1.3 Internal metal surfaces. Internal metal surfaces normally in contact with oil shall be coated with grade 30 of MIL-L-21260 (P-10) preservative, and those normally in contact with water shall be coated with grade 5 of MIL-C-16173. Application may be by flushing, fogging, or spraying, avoiding excessive use of preservatives. After thoroughly draining, the threads of drain plugs shall be coated with type P-10 preservative and reinserted. When plugs are not provided, openings shall be closed with plugs or caps conforming to MIL-C-5501 or with tape conforming to PPP-T-60, type IV.

3.6.1.1.4 Assembled equipment. Each assembled piece of equipment shall be preserved as a complete unit in accordance with the applicable method of MIL-P-116 as specified in Table I. Preservation requirements for equipment not covered by Table I shall be as specified by the procurement officer (see 6.2).

3.6.1.1.5 Engines. Engines shall be preserved in accordance with level A requirements of MIL-E-10062 and the applicable preparation types and methods therein.

3.6.1.1.6 Electrical component parts. Electrical component parts and motors shall be preserved in accordance with level A requirements of MIL-E-16298 and MIL-E-17555, as applicable.

3.6.1.1.7 Technical publications. Unless otherwise specified (see 6.2), technical publications and handbooks shall be preserved in accordance with the requirements for method IC-1 and secured to the basic unit.

3.6.1.1.8 Sets, kits, and outfits w/carrying case. Item(s) and equipment furnished in sets, kits, and outfits shall be individually preserved per Table I, wrapped and placed in slots or contour cutouts provided in the carrying case, or shall be arranged within the set or kit container in a compact manner. Cushioning shall be provided to prevent movement and damage during shipment, storage, and handling.

3.6.1.2 Unit pack quantity and container. The unit pack quantity shall be as specified (see 6.2). Unless otherwise specified or required by the preservation method (see 6.2), the unit containers shall conform to one or more of the following specifications.

| | |
|---|---|
| UU-S-48 | - Sacks, Shipping, Paper |
| PPP-B-15 | - Bags and Envelopes, Cellophane, for Packaging |
| PPP-B-35 | - Bags, Textile-Shipping, Burlap, Cotton and Waterproof Laminated |
| PPP-B-566 | - Boxes, Folding, Paperboard |
| PPP-B-636 | - Boxes, Shipping Fiberboard |
| PPP-B-676 | - Boxes, Setup |
| PPP-B-1364 | - Box, Corrugated Fiberboard, High-Strength, Weather-Resistant, Double-Wall |
| PPP-C-96 | - Cans, Metal, 28 Gage and Lighter (Type V, Class 1 or 2) |
| PPP-D-723 | - Drums, Fiber |
| MIL-B-117 | - Bags, Sleeves and Tubing-Interior Packaging |
| MIL-C-3955 | - Cans, Fiber, Spirally Wound |
| MIL-B-26701 | - Bottles, Screw Cap and Carboys Polyethylene Plastic |
| MIL-B-43014 | - Boxes: Water-Resistant Paperboard; Folding, Set-up and Metal-Stayed |

When the unit container is also the shipping container, the containers of 3.7 shall apply. Carrying cases may serve as the unit container; however, when used as the inner container of a specific method of packaging, all corners, hinges, and latches shall be cushioned to prevent rupturing of barrier material. Sets, kits, and



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

outfits furnished in canvas, leather, or plastic shall be unit packaged submethod IC-3. Sets, kits, and outfits furnished in metal or wood cases not exceeding ten pounds shall be unit packaged submethod IC-2; those exceeding ten pounds shall be unit packaged method I or III, as applicable. All seams, corners, and joints shall be waterproof sealed with tape in accordance with the appendix to PPP-B-636. When specified (see 6.2), unit containers not conforming to weather-resistant requirements shall be overpacked or overwrapped with weather-resistant boxes or barrier material as applicable. Closure of containers shall be in accordance with the applicable container specification and the appendix thereto.

3.6.1.2.1 Transparent plastic packages. When specified (see 6.2), transparent plastic packages conforming to methods IC-7 or IC-9 may be utilized in lieu of the methods specified in Table I herein. Shaped, preformed, or molded packages shall incorporate provisions for separating unit quantities by scoring or other media. The plastic shall be compatible with other material utilized in the process of preservation.

3.6.1.3 Cushioning and cushioning materials. Item(s) will be cushioned as required for the method of preservation in accordance with the guidelines and criteria of MIL-P-116. Cushioning materials used within the unit package shall conform to PPP-P-291, PPP-C-795, or PPP-C-843. Other cushioning materials may be utilized provided they conform to the cushioning requirements of MIL-P-116 and are noncorrosive to the item(s) packaged. Excessive use of cushioning material shall be avoided.

3.6.1.4 Intermediate packs. When specified (see 6.2), unit packs shall be intermediate packaged in containers conforming to PPP-B-636, grade W5c or V3c, or MIL-B-43014. Unless otherwise specified, intermediate quantities shall be governed by the size and weight limitations of the applicable container specification. Cushioning shall be provided as necessary to control movement and damage within the container. Closure shall be in accordance with the container specification and the appendix thereto.

3.6.1.4.1 Air Force. Intermediate pack, quantity, and container for Air Force shall be in accordance with MIL-STD-794.

3.6.2 Level B. Preservation for level B shall be as specified on detailed packaging standards or detailed packaging instructions provided by the procuring activity (see 6.2).

* 3.6.3 Industrial preservation. Industrial preservation of welding and soldering equipment shall conform to ASTM D3951-82.

* 3.7 Packing. Packing shall be level A, B, or Industrial, as specified (see 6.2).

* 3.7.1 Level A. Unless otherwise specified (see 6.2), welding and soldering equipment, supplies, and accessories preserved as specified (see 3.6), shall be packed in shipping containers conforming to the following specifications.

| | |
|---|---|
| QQ-S-781 | - Strapping, Steel, and Seals |
| PPP-B-585 | - Boxes, Wood, Wirebound (Class 3 Use) |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood (Overseas Type) |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner (Class 2) |
| PPP-B-636 | - Boxes, Shipping, Fiberboard (Type CF, Class Weather-Resistant, Grade V2s) |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| PPP-B-640 | - Boxes, Fiberboard, Corrugated, Triple-Wall (Class 2, Style E or G) |
| PPP-D-723 | - Drums, Fiber |
| MIL-C-104 | - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted |
| MIL-C-3774 | - Crates, Wood; Open 12,000- and 16,000-Pound Capacity |
| MIL-P-15011 | - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry |
| MIL-C-52950 | - Crates, Wood, Open and Covered |

Selection of container shall be governed by the size and weight limitations of the applicable container specification. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, type I or IV, finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 inches by 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. Unless otherwise specified (see 6.2), trailer mounted welding equipment shall be shipped mobile (unboxed).

Arrangement of contents, cushioning, anchoring, blocking, and bracing shall be in accordance with MIL-STD-1186. Closure and strapping of shipping containers shall be in accordance with the applicable container specification and the appendix thereto.

When specified (see 6.2), intermediate fiberboard boxes conforming to PPP-B-636, type CF, class weather-resistant, closed and sealed as specified in appendix thereof, may be used as the shipping container.

3.7.2 Level B. Unless otherwise specified (see 6.2), welding and soldering equipment, supplies, and accessories preserved as specified (see 3.6), shall be packed in shipping containers conforming to the following specifications.

| | |
|---|---|
| QQ-S-781 | - Strapping, Steel, and Seals |
| PPP-B-576 | - Boxes, Wood-Cleated, Veneer, Paper Overlaid |
| PPP-B-585 | - Boxes, Wood, Wirebound (Class 1 or Class 2 Use) |
| PPP-B-591 | - Boxes, Shipping, Fiberboard, Wood-Cleated (Class II) |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood (Domestic Type, Style A, B, I, or J) |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner (Class 1) |
| PPP-B-636 | - Boxes, Shipping, Fiberboard (Type CF, Class Weather-Resistant, Grade V3c) |
| PPP-B-640 | - Boxes, Fiberboard, Corrugated, Triple-Wall (Class 2) |
| PPP-D-723 | - Drums, Fiber |
| MIL-C-104 | - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted |
| MIL-P-15011 | - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry |
| MIL-C-52950 | - Crates, Wood, Open and Covered |

Selection of container shall be governed by the size and weight limitations of the applicable container specification. Closure and strapping shall be the same as Level A. Unless other specified (see 6.2), trailer mounted welding equipment shall be shipped mobile (unboxed).

Arrangement of contents, cushioning, anchoring, blocking, and bracing shall be in accordance with MIL-STD-1186. Closure and strapping of shipping containers shall be in accordance with the applicable container specification and the appendix thereto.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

* 3.7.3 Industrial packing. Packing of welding and soldering equipment shall conform to ASTM D3951-82.

3.7.4 Unitized and containerized loads. When specified (see 6.2), items packed as specified in 3.7 shall be unitized or containerized in accordance with MIL-STD-147.

3.8 Marking.

3.8.1 Military. When military levels of preservation and packing are specified, marking shall be in accordance with MIL-STD-129. Special marking required shall be as specified in the contract or order (see 6.2).

3.8.2 Industrial marking. When commercial preservation and packing is specified, marking shall be in accordance with ASTM D3951-82. Special marking required shall be as specified in the contract or order (see 6.2).

3.9 Workmanship. Workmanship shall be such that when the proper procedure is followed, materiel and equipment being processed will receive the maximum protection against corrosion, deterioration, and damage during handling, shipment, and storage.

3.10 Performance.

3.10.1 Rough handling. When specified (see 6.2), the completed pack shall provide protection from rough handling sufficient to prevent damage to the packaged item(s) when the pack is subjected to the rough handling tests of MIL-P-116. Following the tests, the packaged item(s) shall show no evidence of damage, and the exterior container shall not reveal any failure caused by improper application or design of interior cushioning, blocking, or bracing. All barriers, wrappings, interior containers, blocking, bracing and anchoring shall be intact and capable of providing the intended protection.

3.10.2 Exposure. When specified (see 6.2), the completed pack shall provide protection from exposure to the elements sufficient to prevent damage to the packaged item(s) when the pack is subjected to the cyclic exposure test of MIL-P-116. Following the test, the tested package, when opened, shall show no evidence of moisture nor shall the packaged item show corrosion.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

4.2.1 First article inspection. First article inspection is required by the contract or purchase order (see 3.1), the contractor shall subject the first article pack to the applicable first article inspection and tests specified in MIL-P-116 and the examinations in the applicable container specification and Appendix B of MIL-STD-1186 to determine conformance to the requirements of this specification. When specified (see 6.2), first article packages shall be subjected to the rough handling and cyclic exposure tests(s) (one or both) as specified in the first article inspection requirements of MIL-P-116. Failure of the first article to pass any of the inspections or tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production packs offered for acceptance under the contract or purchase order.

4.2.2.1 Level A and B preservation and packing. Quality conformance inspection shall consist of the applicable quality conformance inspections and tests specified in MIL-P-116 and the examinations in the applicable container specification and Appendix B of MIL-STD-1186 to determine conformance to the requirements of this specification. When specified (see 6.2), the packs shall be subjected to the rough handling and cyclic exposure test(s) (one or both) as specified in the quality conformance inspection requirements of MIL-P-116. Failure of the packs to pass any inspection or test shall be cause for rejection.

4.2.2.2 Industrial preservation and packing. Industrial preservation and packaging shall be examined to determine conformance with the requirements of ASTM D3951-82. Failure of the packs to pass the examination shall be cause for rejection.

4.3 Sampling.

4.3.1 Level A and B preservation and packing. Sampling for quality conformance inspection shall be conducted in accordance with MIL-STD-105 at the inspection level and acceptable quality level (AQL) specified in MIL-P-116 and Appendix B of MIL-STD-1186, respectively. Sampling for examination of the container shall be at level II and an AQL of 4.0. When specified (see 6.2), sampling for the rough handling and cyclic exposure test(s) (one or both) shall be at inspection level S-2 with an AQL of 4.0.

4.3.2 Industrial preservation and packing. Sampling for quality conformance inspection shall be conducted in accordance with MIL-STD-105 at inspection level S-4 with an AQL of 4.0.

4.3.3 Disposition of samples. All samples used for inspection and testing shall be reprocessed as necessary. They may, after reprocessing in accordance with the original method of preservation, be considered a part of the original lot. When the packaged item may have been damaged as a result of testing, the item shall be inspected as necessary to determine its acceptability.

5 PACKAGING

5.1 This section is not applicable to this specification.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

## 6 NOTES

6.1 Intended use. This specification is intended to be used for reference in section 5 of commodity specifications or direct reference in contracts or orders. The cleaning, drying, preservative application, packing, and marking requirements specified herein are intended to insure proper and safe storage and transportation of welding and soldering equipment, supplies, and accessories for direct shipment to Government activities or shipments to be processed at a military activity or agency.

6.2 Ordering data. Purchasers should exercise any desired options offered herein. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. First article, when required (see 3.1).
c. Special cleaning processes or drying procedures if required (see 3.5).
d. Specify level A or B, or industrial preservation (see 3.6).
e. When a preservative other than that specified is required (see 3.6.1.1).
f. Method of preservation if other than specified (see 3.6.1.1 and 3.6.1.2.1).
g. When VCI treated materials are to be used as a preservative (see 3.6.1.1.1).
h. When a contact preservative is required with VCI (see 3.6.1.1.1).
i. The method of preservation for assembled equipment not listed in Table I (see 3.6.1.1.4).
j. Packaging of technical publication if different (see 3.6.1.1.7).
k. Unit pack quantity and unit container to be utilized if different (see 3.6.1.2).
l. If unit container is to be overpacked or overwrapped (see 3.6.1.2).
m. When unit pack require intermediate packaging (see 3.6.1.4).
n. Specify level A or B, or industrial packing (see 3.7).
o. When trailer mounted welding equipment is to be boxed (see 3.7.1, 3.7.2 and 3.7.3).
p. Selection of shipping container to be utilized if different (see 3.7.1 and 3.7.2).
q. When intermediate container may become the shipping container (see 3.7.1).
r. When item(s) are to be unitized or containerized (see 3.7.4).
s. Special marking if required (see 3.8.1 and 3.8.2).
t. Rough handling test, if required (see 3.10.1, 4.2.1 and 4.2.2.1).
u. Cyclic exposure test, if required (see 3.10.2, 4.2.1 and 4.2.2.1).
v. Specified detailed packaging standards or instructions (see 3.6.2).

6.2.1 Level B preservation and packing. Level B preservation and packing is intended to provide adequate protection against known conditions that are less hazardous than Level A is designed to meet. This level provides a higher degree of protection than afforded by Commercial. The design of Level B is based on firmly established knowledge of shipment, handling, and storage conditions to be encountered, and on the determination that the cost of preparation is less than for Level A (see 3.6.2).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

* 6.4 Disposability methods. Environmental Pollution Preventative Measures are contained in the materials' specifications indicated herein. Refer to the materials' specifications (or preparing activity) for recommended disposability methods.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - SM
Navy - SH, YD
Air Force - none
DSA - IP

Preparing Activity:
Army - AL

Project No. PACK-0707



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Preservation-packaging of soldering and welding supplies and accessories.

| ITEM | Preservation-Packaging, methods of MIL-P-116 | Applicable type of Preservative MIL-P-116 | Special Remarks |
|---|---|---|---|
| COLUMN 1 | COLUMN 2 | COLUMN 3 | COLUMN 4 |
| ADAPTER | | | |
| Acetylene Hose to Reg. | III | | PPP-B-566 |
| Oxy. Acety. Fitting | III | | PPP-B-566 |
| Propane Fitting | III | | PPP-B-566 |
| Regulator | III | | PPP-B-566 |
| Tip | III | | PPP-B-566 |
| Tube to Hose | III | | PPP-B-566 |
| APRON | | | |
| Welders | IC-3 | None | |
| BAG | | | |
| Canvas | IC-3 | None | |
| Torch Equipment | IC-3 | None | |
| Welding Rods | IC-3 | None | |
| BALL | | | |
| Relief Valve | III | | |
| BAR | | | |
| Clamp Tool Post | I | P-1 | |
| BATTERY | | | |
| Storage | | | See MIL-B-208 |
| BODY | | | |
| Fixture Tool Post | I | P-2 | |
| Welding Torch | IC-2 | None | |
| BOX | | | |
| Tool | I or III | P-1 or P-19 | See 3.6.1.1.2 |
| BRAZING & SOLDERING | | | |
| Set | IC-2 | None | See 3.6.1.2 |
| BRAZING ALLOYS | | | |
| Copper (Bar) | III | | |
| Strip | III | | |
| Wire | III | | |
| Silver (Strip) | III | | |
| Wire | III | | |
| BRUSH | | | |
| Wire | III | | PPP-B-566 |
| CABLE ASSEMBLY | I or III | None | |
| CABLE LUGS | I or III | None | |
| CAP | I | P-2 | |
| Plastic | III | | |
| CARBON BLOCK | III | | |
| CARBON PASTE | III | | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Preservation-packaging of soldering and welding supplies and accessories (continued).

| COLUMN 1 | COLUMN 2 | COLUMN 3 | COLUMN 4 |
|---|---|---|---|
| CARBON ROD | III | | |
| CARTRIDGE | III | | |
| CASE | | | |
| Carrying | I or III | P-1 | See 3.6.1.1.2 |
| Gun | I or III | P-1 | See 3.6.1.1.2 |
| CHAMBER | | | |
| Precombustion | IC-1 | None | MIL-B-117 |
| CHUCK | I | P-1 | |
| CLAMP | | | |
| Pipe Aligning | I | P-10 | |
| Ground | I or III | P-1 | |
| CLEANER | | | |
| Tip | III | | MIL-B-117 |
| CLEANER SET | III | | PPP-B-566 |
| CLIP | | | |
| Electric Sold. Iron | I or III | P-1 | See 3.6.1.1.2 |
| CONNECTORS (Cable) | I or III | | |
| CUTTING ATTACHMENT | IC-1 | None | MIL-B-117 |
| CUTTING & BEVELING MACHINE | II | | See 3.6.1.1.4 |
| CUTTING MACHINE | II | | See 3.6.1.1.4 |
| CYLINDERS | I or III | P-18 | VCI on fitting |
| Torch Set | I or III | P-18 | VCI on fitting |
| DIAPHRAGM | | | |
| Regulator, Oxy. | IC-1 | None | MIL-B-117 |
| Rubber | IC-1 | None | MIL-B-117 |
| Valve | IC-1 | None | MIL-B-117 |
| ELECTRODE (Welding) | | | See MIL-W-10430 |
| Carbon | III | | See O-S-801 |
| ELECTROLYTE | | | |
| FLUX SOLDERING | | | |
| Granular | III | | |
| Liquid | III | | |
| Paste | III | | |
| Stick | III | | |
| FLUX WELDING | | | |
| Power | III | | |
| Granular | III | | |
| FLUX RECOVERY MACHINE | IA-15 | P-1 & P-2 | |
| FURNACE | | | |
| Flux Welding | IIa | None | See 3.6.1.1.4 |
| Soldering | IIb | None | See 3.6.1.1.4 |
| Metal Melting | I | P-1 & P-6 | See 3.6.1.1.4 |
| GASKET | III | | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Preservation-packaging of soldering and welding supplies and accessories (continued).

| COLUMN 1 | COLUMN 2 | COLUMN 3 | COLUMN 4 |
|---|---|---|---|
| GLAND | | | |
| Hose Connection | III | | |
| GLOVES | | | |
| Leather | IC-3 | None | |
| Canvas | IC-3 | None | |
| GOGGLES | III | None | |
| HAMMER (Welders) | I | P-2 | |
| HANDLE | | | |
| Torch | IC-2 | None | |
| Sold. Iron Electric | IA-8 | None | |
| HEAD | | | |
| Mixing | IC-1 | None | |
| Soldering | IC-1 | None | |
| Welding | IC-1 | | |
| HEATING ELEMENT | | | |
| Electric | IA-8 | None | |
| HEAT SINK | III | | |
| HELMET (Welders) | III | None | |
| HOLDER | | | |
| Tool | III | | |
| Electrode | III | | |
| Welding Rod | III | | |
| Soldering Iron | I or III | P-1 | See 3.6.1.1.2 |
| HOSE | | | |
| Oxygen | | | See MIL-H-775 |
| Acetylene | | | See MIL-H-775 |
| Air | | | See MIL-H-775 |
| HOSE ASSEMBLY | IC-2 | | |
| IGNITER | IC-1 | None | The flint shall be unscrewed (removed) and secured on the hood with PPP-T-60 tape |
| INSULATOR BRUSH CLAMP | IC-1 | None | |
| JAWS | | | |
| Electrode Holder | III | | |
| KIT (Wire Cleaning) | III | | |
| LEAD ELECTRICAL | IC-1 | None | |
| LEAD SETS | | | |
| Soldering and Brazing | IC-2 | None | |
| LENSES | | | |
| Goggle | III | None | |
| Helmet | III | None | |
| LEVER (Valve Cutting) | I | P-1 | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Preservation-packaging of soldering and welding supplies and accessories (continued).

| COLUMN 1 | COLUMN 2 | COLUMN 3 | COLUMN 4 |
|---|---|---|---|
| LINER (Conduit) | III | | |
| MACHINE METALIZING | | | |
| Fuse Bond Metal Spraying | IIb | None | See 3.6.1.1.4 |
| MANIFOLD | | | |
| Gas cylinder | III | | |
| METALIZING OUTFIT | I | P-18 | See 3.6.1.1.8 |
| MITTENS | IC-3 | None | |
| MOISTURE STABILIZER | IC-2 | P-1 | See 3.6.1.1.2 |
| NIPPLE OUTFIT | III | | |
| NOZZLE | III | | |
| NUT | | | |
| Mixing Head | III | | |
| Union | III | | |
| OILER | I | P-10 | |
| PACKING | IC-1 | None | |
| PIPE (Torch Extension) | IC-1 | None | |
| PILOT (Regulator Seat) | IC-1 | None | |
| PIN SET | IC-1 | None | |
| PIN (Valve Lever Cut) | IC-1 | None | |
| PLIERS (Slip Joint) | I | P-2 | |
| PLUG COUPLER | I | P-2 | |
| POSITIONERS | I | P-1 & P-2 | See MIL-E-16298 (Alternate method) |
| POWDER | | | |
| Welding | III | | |
| POWER UNIT | IA-15 | None | See MIL-E-17555 |
| PUNCH | I | P-2 | |
| RACK | | | |
| Welding Rod | I or III | P-1 | See 3.6.1.1.2 |
| Cylinder | III | | |
| RECEPTACLE | I or III | None | |
| REEL (Wire) | I or III | P-1 | See 3.6.1.1.2 |
| Hose (Welding) | I or III | P-1 or P-19 | See 3.6.1.1.2 |
| REGULATOR (Pressure) | IC-2 | None | |
| REGULATOR FILTER UNIT | IC-2 | None | |
| RESEATING TOOL | | | |
| Cutting Tip Torch | I | P-2 | |
| Needle Welding Torch | I | P-2 | |
| Welding Tip Torch | I | P-2 | |
| RETAINER (Seat) | IC-1 | None | |
| RING | | | |
| Packing | IC-1 | None | |
| Slip Diaphragm | IC-1 | None | |
| Snap Coupler Plug | IC-1 | None | |
| RODS (Welding) | | | See MIL-W-10430 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Preservation-packaging of soldering and welding supplies and accessories (continued).

| COLUMN 1 | COLUMN 2 | COLUMN 3 | COLUMN 4 |
|---|---|---|---|
| SCREEN (Welders) | III | None | |
| SOLDER | | | |
| Bismuth Lead Alloy | III | | |
| Lead Alloy | | | |
| Bar | III | | |
| Ingot | III | | |
| Ribbon | III | | |
| Strip | III | | |
| Wire | III | | |
| Lead-Tin Alloy | | | |
| Bar | III | | |
| Wire | III | | |
| Tin Alloy | | | |
| Bar | III | | |
| Wire | III | | |
| SOLDERING AID | | | |
| Circuit Tester | IA-8 | None | |
| SOLDERING & BRAZING OUTFIT | IC-2 | None | See 3.6.1.1.8 |
| SOLDERING FORK | IA-15 | None | |
| SOLDERING GUN | IA-15 | None | |
| SOLDERING IRON | | | |
| Electric | IA-15 | None | |
| Non-Electric | III | | |
| SOLDERING PENCIL | IA-8 | None | |
| SOLDERING PLIERS | IA-15 | None | |
| SWITCH (Foot Control) | IC-2 | None | |
| TABLE (Welding) | I or III | P-1 | See 3.6.1.1.2 |
| TIP | | | |
| Igniter | IC-1 | None | |
| Blow Torch | III | | |
| Brazing & Soldering | III | | |
| Oxygen Acetylene | IC-1 | None | |
| Cutting | IC-1 | None | |
| Cutting Torch | IC-1 | None | |
| Soldering Gun | IC-1 | None | |
| Soldering Iron, Elec. | IC-1 | None | |
| Soldering Iron, Non-Elec. | III | None | |
| TIP SET | IC-1 | None | |
| TOOL (Shaft) | I | P-2 | |
| TOOL KIT (Welding) | I | P-18 | See 3.6.1.1.8 |
| TOOL SET (Metal Spray) | I | P-18 | See 3.6.1.1.8 |
| TORCH (Acetylene) | IC-2 | None | |
| Arc-Oxygen Cutting | IC-2 | None | |
| Cutting | IC-2 | None | |
| Welding | IC-2 | None | |

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Preservation-packaging of soldering and welding supplies and accessories (continued).

| COLUMN 1 | COLUMN 2 | COLUMN 3 | COLUMN 4 |
|---|---|---|---|
| TORCH EQUIPMENT | I | | See 3.6.1.1.8 |
| TORCH OUTFIT | IC | | See 3.6.1.1.8 |
| TORCH SET | IC | | See 3.6.1.1.8 |
| TUBE (Mixer) | III | | |
| UNION ATTACHMENT | III | | |
| VALVE | | | |
| Acetylene | IC-2 | None | |
| Compensating Reg. | IC-2 | None | |
| Needle T-Handle | IC-1 | None | |
| Oxygen | IC-2 | None | |
| Safety Relief | IC-1 | None | |
| VISE | I or III | P-1 | See 3.6.1.1.2 |
| WASHER | | | |
| Jack Insulating | IC-1 | None | |
| Back Cap | IC-1 | None | |
| WELDER | | | |
| Machine Arc, Selenium Rectifier Type | IIa | None | See 3.6.1.1.4 |
| Spot, portable w/stand and 10 Tongs | I & IIb | P-1 & P-2 | See 3.6.1.1.4 |
| WELDING MACHINE, ARC | | | |
| Gasoline engine driven Skid mtd. DC Aux. Generator | IIa | None | See 3.6.1.1.4 |
| DC, power take-off driven | IIa | None | See 3.6.1.1.4 |
| Selenium rectifier type mounted | IIa | None | See 3.6.1.1.4 |
| Gasoline or Diesel Engine driven, trailer mounted | I | P-1, P-2, P-3, P-11, P-19 | See MIL-E-16298 (Alternate method) MIL-STD-281, and MIL-E-10062 |
| WELDING MACHINES | | | |
| 50 pounds and under | IIb | None | See 3.6.1.1.4 |
| Over 50 pounds | IIa | None | See 3.6.1.1.4 |
| WELD PREHEAT SYSTEM | II | None | See 3.6.1.1.4 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander US Army Armament MAterial Readiness Command
Attn: DRSAR-LEE-S
Rock Island IL 61299

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2025-12-27T03:25Z
Check the source to verify that this is the current version before use.